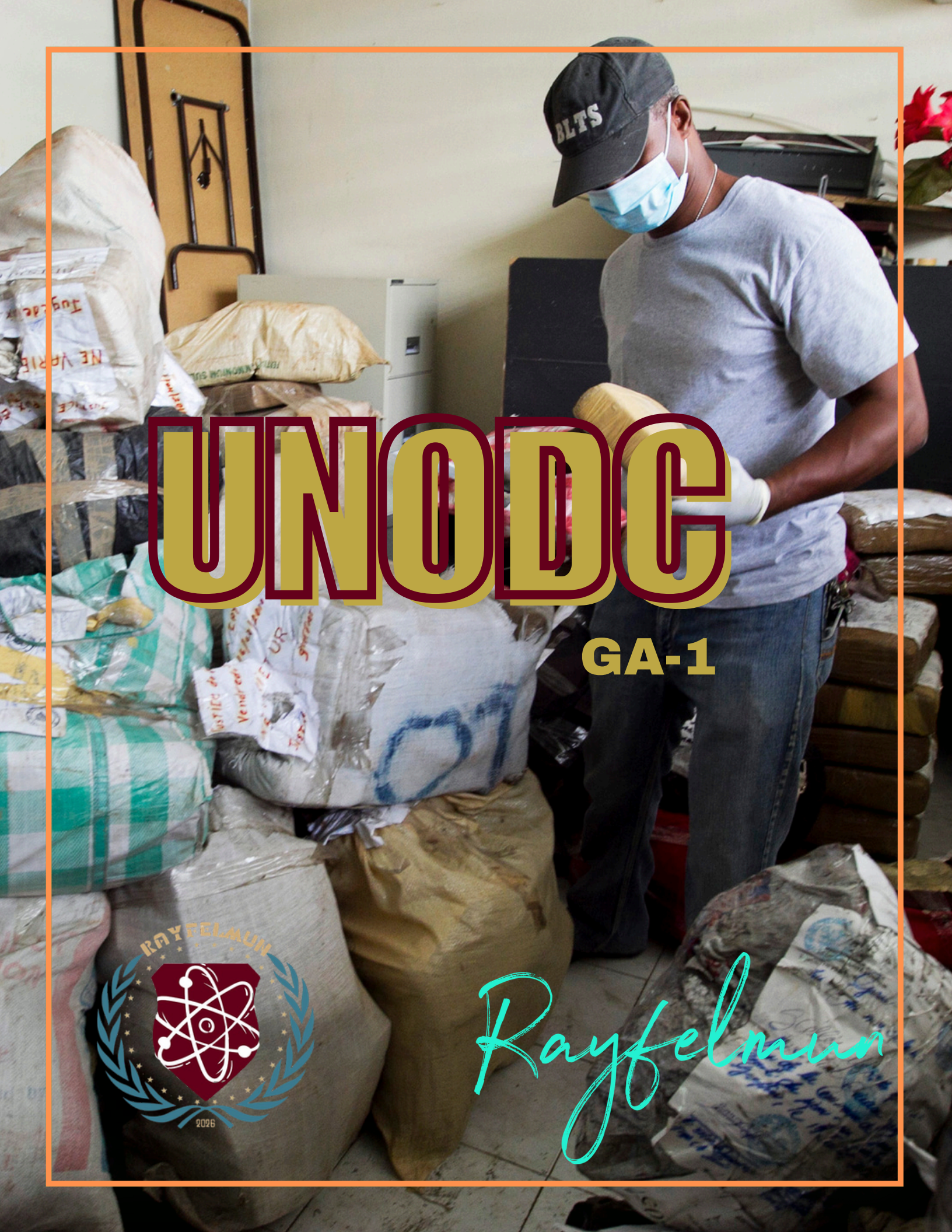




UNODC

GA-1



Rayfelmur

Agenda Item: Strengthening International Standards for Forensic Evidence Collection and Chain of Custody

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1.

Letter From the Head of the Academy

Dear Delegates;

With great enthusiasm, I welcome you all to the second edition of our school's Model United Nations conference! As the Head of the Academy, I am honored to witness the extraordinary talent and dedication each of you brings to this platform for dialogue and collaboration. This year, we are excited to present six dynamic committees, each tasked with addressing some of the most significant global issues.

In the Legal Committee, delegates will engage with one of the most complex and sensitive topics of our time: the legalization of abortion. The issue of abortion has been the subject of intense debate in different parts of the world for many years, shaped around fundamental concepts such as individual bodily autonomy, the right to life, public health, and social responsibility. States have adopted various policies on this issue in line with cultural and legal differences, leading to different practices internationally.

Considering different perspectives within a framework of mutual respect, examining the effects of existing practices, and developing proposals that can provide solutions to the problems facing the international community are the most important elements of our committee. During our discussions, we will comprehensively address issues such as human rights, women's health, societal impacts, and legal regulations.

Thank you for being part of this journey. I wish you thoughtful discussions, meaningful connections, and an inspiring MUN experience. If you have any questions, my door is always open.

Sincere Regards,

Defne Yaman *Head of Academy*

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2.

Letter from the Under Secretary General of UNODC

Dear Delegates,

Welcome to RAYFELMUN '26 and the United Nations Office on Drugs and Crime (UNODC) Committee. My name is Betül Ece Altun, and I am honored to serve as your Under-Secretary-General.

This year, we will discuss “*Strengthening International Standards for Forensic Evidence Collection and Chain of Custody*.” As technology continues to transform criminal investigations, ensuring the integrity and reliability of forensic evidence has become a crucial global challenge. From evidence collection to courtroom presentation, maintaining a secure chain of custody is essential for achieving justice.

During our sessions, delegates will explore topics such as forensic technologies, artificial intelligence, evidence tampering, and international cooperation. I encourage each of you to participate actively, think critically, and work together to develop practical and innovative solutions.

I look forward to meeting all of you and witnessing productive debates throughout the conference. I wish you the best of luck in your preparation and hope you enjoy an unforgettable MUN experience.

Sincerely,

Betül Ece Altun

Under-Secretary-General

UNODC Committee

Key Words and Their Definitions

Chain of Custody: The chronological, written record of the seizure, custody, control, transfer, analysis, and disposition of physical or electronic evidence

Evidence Integrity: The assurance that evidence, whether physical or digital, remains authentic, unaltered, and untampered with from its collection to its presentation in court

Preservation: The act of keeping something as it is, especially to prevent it from being damaged, lost, or destroyed

Contamination: The process by which something becomes dirty, polluted, or mixed with other substances in a way that makes it unreliable

Tampering: The act of touching or changing something without permission, especially in a way that causes damage or makes it unreliable

Documentation: Written or recorded information that provides details or proof about something

Admissibility: The quality of being accepted as evidence in a court of law

Authentication: The process of proving that something is real, true, or genuine

Procedure: A set of actions that is the official or accepted way of doing something

Transparency: The quality of being done in an open way so that people can easily see what actions have been taken

Accountability: The fact of being responsible for what you do and being able to explain your actions

LIMS: Laboratory Information Management System

Forensic Spoliation: the loss, destruction, or material alteration of an object or document that is evidence or potential evidence in a legal proceeding by one who has the responsibility for its preservation.

4. Introduction to the Committee

a. What is UNODC?

The United Nations Office on Drugs and Crime (UNODC) is a United Nations office that was established in 1997 as the Office for Drug Control and Crime Prevention by combining the United Nations International Drug Control Program (UNDCP) and the Crime Prevention and Criminal Justice Division in the United Nations Office at Vienna, adopting the current name in 2002.

The agency's focus is the trafficking and abuse of illicit drugs, crime prevention and criminal justice, international terrorism, and political corruption. It is a member of the United Nations Development Group. In 2022–23, it had an estimated biannual budget of US\$822 million. UNODC has country and regional field offices in 21 locations around the world, where field staff work directly with institutions, civil society organizations and local communities in order to develop and implement drug control and crime prevention programmes that are tailored to the needs of assisted countries. Two Commissions exist within UNODC: the Commission on Narcotic Drugs (CND) and the Commission on Crime Prevention and Criminal Justice (CCPCJ). Furthermore, UNODC guards the international drug- and crime-related treaties such as the Convention on Psychotropic Substances or the Convention against Corruption.

The three pillars of the UNODC work programme are:

- Research and analytical work to increase knowledge and understanding of drug- and crime-related issues and expand the evidence base for policy and operational decisions
- Normative work to assist states in the ratification and implementation of international treaties and the development of domestic legislation on drugs, crime and terrorism
- Field-based technical cooperation projects enhancing the capacity of Member States to counteract illicit drugs, crime and terrorism

UNODC advocates a balanced approach to drug control, emphasising the importance of preventive education and treatment for addicts, as well as measures

to reduce production and trafficking. UNODC's Global Assessment Programme supplies worldwide accurate and current statistics on illicit drug consumption, which is crucial for the successful development of best prevention strategies.

b. History of UNODC

The United Nations Office on Drugs and Crime was officially created in 1997 to strengthen the United Nations' ability to address global problems related to drugs, organized crime, and criminal justice. It was formed through the merger of the United Nations International Drug Control Programme (UNDCP) and the Centre for International Crime Prevention, combining the UN's work on drug control with its efforts to combat international crime. The organization was initially called the Office for Drug Control and Crime Prevention, but in 2002 it was renamed UNODC to better reflect its broader mission. Its headquarters are located in Vienna, where it works closely with international bodies such as the Commission on Narcotic Drugs and other UN agencies.

Over time, UNODC's responsibilities expanded to include fighting human trafficking, corruption, terrorism, cybercrime, and transnational organized crime. The office also supports the implementation of major international treaties such as the United Nations Convention against Transnational Organized Crime and the United Nations Convention against Corruption.

Today, UNODC operates through regional and country offices around the world, providing research, policy guidance, training for law enforcement and judicial systems, and international cooperation programs to help member states strengthen the rule of law and effectively combat global criminal networks.

5. Introduction to the Agenda Item

a. What is Forensic Science?

Derived from the Latin term *forensis* which means a public debate or discussion, forensics in the modern sense implies courts of law. Forensic Science is therefore the application of science, and the scientific method to the judicial system. The important word here is science. A forensic scientist will not only be analyzing and interpreting evidence but also challenged in court while providing expert witness testimony. Forensic science (often shortened to forensics) is used to enforce laws

and government regulations and statutes, to resolve disputes, to assess blame and establish responsibility, and to improve public safety. Because science is now used routinely in litigation, various groups, including lawyers, judges, enforcement officials, and the public, need to know what forensic science can and cannot do. Forensic scientists use innovative scientific techniques to examine and interpret evidence in connection with civil and criminal proceedings. In criminal law, forensics science can help prove the guilt or innocence of the defendant. In civil actions, forensics can help resolve a broad spectrum of legal issues through the identification, analysis, and evaluation of physical evidence.

The traditional disciplines of forensic science include:

- Toxicology (study of alcohol and drugs)
- Serology (study of blood and other biological fluids)
- Questioned document examination (examination of documents, handwriting comparison, study of inks, typewriter imprints, counterfeiting etc.)
- Chemistry
- Firearms identification and ballistics (study of marks and striations on bullets)
- Hair and fibre analysis
- Pathology
- Odontology (study of bite marks, teeth structure)
- Anthropology and the determination of a biological profile

b. What is Chain of Custody?

When evidence is presented in a courtroom, it is imperative that the integrity of the evidence has not been compromised. In criminal trials, the standard of proof for conviction is beyond a reasonable doubt, meaning that no rational or prudent person would believe that the Defendant is innocent. It is the highest burden of proof required in law, as in civil cases the standard of proof is only preponderance of the evidence, meaning that the evidence points to the Defendant being guilty. These differences make sense because the penalties handed out in criminal cases

are more than just monetary. Would it be lawful or ethical for a jury to sentence a murderer to the death penalty or life in prison on compromised evidence? Absolutely not. The chain of custody allows investigators and jurors to be confident in the safety and integrity of evidence.

So, what is the chain of custody? The definition of the chain of custody is the procedure, documentation, tracking, and protecting of evidence on its journey from a crime scene to the courtroom. This documentation process protects evidence from contamination and tampering. It is also sometimes called the chain of evidence. If the chain of custody has been broken or tampered with, evidence can be ruled as inadmissible, meaning it cannot be presented or believed to be authentic in a court of law.

c. Certain Significant Nations

China

China has been investing time, energy and resources into forensic science since the 1980's and globally-renowned forensic scientist Henry Chang-yu Lee believes it's about to pay off tipping China to become a world leader in high-tech evidence collection.

"I believe the technology in China will be more advanced than ever in the United States within five years," the Chinese-American expert said in a recent interview with China Daily. Lee, who has racked up more than five decades of experience in forensic science, has worked on a number of high-profile criminal cases in the US, but has also shared his wealth of knowledge with students, lawyers, judges and law enforcement in China over the years. "The apparatus and devices used to identify fingerprints or footprints, for example, were very simple when I first visited Chinese forensic laboratories," he said. However, he has seen the technology improve over the years and there have been many advances, particularly in electronic evidence collection and fraud prevention by means of real-time monitoring.

In 2016, Lee and several other experts established the Silk Road Forensic Consortium in Xi'an, Shaanxi province, to fight crime and safeguard security by boosting scientific exchanges among countries involved in the Belt and Road Initiative.

The consortium, which has 150 members from 30 countries and regions, provides an open platform for forensic specialists, police officers and judges to share ideas and difficulties as well as experiences in DNA identification studies. Lee, who acts as chairman, said, "Although we speak different languages in our daily lives, we all speak the same 'language' at work, and that's the language of the criminal investigation. We share the same goal – to speak for the dead using forensic science." In September, at the organisation's third annual conference in Yantai, Shandong province, Lee announced plans to unify DNA identification standards among its members to try and build a mutual DNA database that can better solve criminal cases.

Germany

Police in the German state of Bavaria have the power to use forensic DNA profiling after a controversial law passed in 2018 in the Landtag, the state parliament in Munich. The law was the first in Germany to allow authorities to use DNA to help determine the physical characteristics, such as eye colour, of an unknown culprit. The new DNA rules are part of a broader law which has drawn criticism of the wide surveillance powers it gives the state's police to investigate people they deem an "imminent danger," people who haven't necessarily committed any crimes but might be planning to do so. The move was prompted, in part, by the rape and murder of a medical student in Freiburg, Germany, in late 2016. An asylum seeker, originally from Afghanistan, was convicted of the murder and sentenced to life in prison. But some authorities complained that they could have narrowed their search more quickly if they had been able to use trace DNA to predict what the suspect would look like. Federal and state laws previously only allowed investigators to use DNA to look for an exact match between crime scene evidence and a potential culprit, either in a database of known criminals or from a suspect.

United Kingdom

Prior to its closure in 2012, the UK Forensic Science Service (FSS) was a world-leader in forensic technology. It pioneered the use of the handheld breath alcohol roadside tester, and the DNA national database was first worked on and initially tested on all staff and police forces to ensure its reliability. The organisation later pioneered the use of large-scale DNA profiling for forensic identification and crime detection when it moved the facilities to Birmingham. This enabled the launch of the world's first DNA database on 10 April 1995. The FSS's innovative and extremely sensitive DNA profiling technique called LCN (low copy number) was used in convicting Antoni Imiela (the M25 rapist).

As well as Ronald Castree (for the murder of Lesley Molseed in 1975) but the organisation came under attack when it failed to recover blood stains from a shoe in the murder of Damilola Taylor. Forensic laboratories in the UK are now privately-owned but are experiencing similar financial difficulties, a recent inquiry by the House of Lords heard. Mark Pearse, the commercial director in the forensics division of Eurofins, one of the three major providers in the UK, described an “unsustainable toxic set of conditions” when he appeared before the inquiry. Representatives from the two other largest providers – Key Forensics, which had to be bailed out by police last year after going into administration, and Cellmark – raised similar concerns. However, that is not to say that the UK is not involved in researching and implementing new forensic technologies. Forensic scientists are currently working with the British military to open the United Kingdom's first body farm — a site where researchers will be able to study the decomposition of human remains.

Details are not yet finalized, but the plans are at an advanced stage: project leaders hope this year to open the farm, also known as a forensic cemetery or taphonomy facility, after the discipline devoted to the study of decay and fossilization. Such sites generate data on tissue and bone degradation under controlled conditions, along with chemical changes in the soil, air, and water around a corpse, to help criminal and forensic investigators.

Netherlands

The Netherlands Forensic Institute (NFI) is one of the world's leading forensic laboratories. From its state-of-the-art, purpose-built premises in The Hague, the NFI provides products and services to a wide range of national and international clients.

To ensure that their work remains at the forefront of developments, the Netherlands Forensic Institute invests heavily in Research and Development. In this way, it lays the foundations for innovative forensic methods and technologies that will play an important part in the coming decades. Amongst these innovative forensic technologies is the invention of Hansken, a system that can store large quantities and diverse data easily from different sources. All data is stored, indexed, enriched and made rapidly searchable, cutting down the turnaround time of forensic evidence. It now contains over 150 samples of glass from a large number cases. In several cases, this glass database has linked suspects to several crimes.

Offenders who carry out robberies, smash-and-grab raids or ARM gas attacks may have splinters of glass on their clothes or in the soles of their shoes and these splinters of glass can remain in place for months, even though they are barely visible to the naked eye, if at all. These splinters can be of great value. The composition of each piece of glass is unique because of minuscule contaminants in the raw materials for making glass. By comparing the unique composition of splinters of glass found on a suspect to glass from the database, it is possible to check whether that glass originates from a crime committed earlier. The glass analysts of the NFI measure the concentration of twenty elements in each piece of glass. This produces a kind of 'chemical fingerprint' of the material.

United States of America

It will come as no surprise that at the forefront of cutting-edge forensic technology is the USA, home to over 400 crime labs and the biggest crime lab in the world, the FBI Laboratory.

To help train government and industry organisations on cyberattack prevention, as part of a research project for the U.S. Army, scientists at The University of Texas at San Antonio, have developed the first framework to score the agility of cyber attackers and defenders. “The DOD and U.S. Army recognize that the cyber domain is as important a battlefield as ground, air and sea,” said Dr. Purush Iyer, division chief, network sciences at Army Research Office, an element of the Army Futures Command’s Army Research Laboratory. “Being able to predict what the adversaries will likely do provides opportunities to protect and to launch countermeasures. This work is a testament to successful collaboration between academia and government.” The framework developed by the researchers will help government and industry organizations visualize how well they out-maneuver attacks. Their work is published in IEEE Transactions on Information Forensics and Security, a top journal for cybersecurity.

Education and training programs in the field of forensics are also on the rise, supported by organisations such as The Forensic Sciences Foundation and the American Academy of Forensic Sciences. In fact, there are 485 Forensic Science schools in the US, so it’s no wonder that it is the home of some of the most influential forensic scientists, such as Dr. Michael M. Baden and Ellis R. Kerley, and is sure to produce a great deal more talent in the future.

6. Liability and Accountability in Cases of Forensic Error or Manipulation

Wrongful conviction, or the conviction of a person for a crime that they did not commit, is one of the greatest travesties of the criminal justice system. As of 2023, The National Registry of Exonerations has recorded over 3,000 cases of wrongful convictions in the United States. Organizations such as The Innocence Project

work to free the innocent and prevent these convictions, so far exonerating 375 people, including 21 who served on death row.

Dr. Jon Gould of the University of California at Irvine has claimed that faulty forensic science is partly to blame for some of these convictions. As one of the architects of research that assesses the impact of forensic science on wrongful convictions, he has cited flawed eyewitness identification, confessions, testimony, police and prosecutorial conduct, defense lawyering, and forensic science as factors related to wrongful convictions. He stressed these are “factors” rather than “causes” because in prior analyses researchers have not been able to draw conclusions about causation because the studies did not use control groups. Forensic misconduct is unethical, inept, or unprofessional actions by expert witnesses or forensic scientists that compromise the integrity of forensic evidence. These actions can be simple procedural mistakes, relying on outdated science, bringing unconscious bias into the process, or deliberately manipulating evidence.

Some of the most common types of forensic misconduct include:

- ~Altering or fabricating physical evidence or test results
- ~Giving testimony that exaggerates the strength or detail of evidence
- ~Using flawed, outdated, or discredited forensic methods
- ~Mishandling, failing to secure, or contaminating crime scene evidence
- ~Failing to disclose evidence that would support your innocence

Some forensic misconduct occurs in the laboratory. Other types occur with evidence handling at the crime scene. In some cases, forensic misconduct means not establishing a controlled chain of custody—knowing who always had access to the evidence. If the chain of custody is incomplete or open, the evidence cannot be trusted. Finally, forensic misconduct can occur in the courtroom, where expert testimony may be misleading. This often means that the experts fail to disclose exculpatory evidence: evidence that casts doubt on the results or points toward your innocence.

Forensic misconduct can easily cause wrongful convictions. Of all the evidence presented at trial, the veneer of scientific validity often makes forensic evidence seem strongest. Forensic experts and other expert witnesses can play on people's trust of science to deceive jurors. This is even more effective when jurors are not scientists themselves. It can make jurors believe there is certainty in a case where there is actually reasonable doubt about a person's guilt. Common forensic misconduct that can lead to wrongful convictions includes evidence tampering. In order to avoid tampering, there must be an unbroken chain of custody from when law enforcement officers gathered the evidence until it is tested. This shows that the evidence was always controlled in a situation that would make tampering difficult.

Even if the evidence itself was not tampered with, forensic misconduct might report inaccurate or misleading test results.

7. The Role of Forensic Technologies in Evidence Collection

For decades, the core of forensic science relied on meticulous human observation and proved analog methods. Today, we are in a revolutionary era. The rapid advancement of forensic technology is transforming every facet of evidence analysis, from the crime scene to the courtroom. Digital tools, artificial intelligence, and sophisticated software are no longer the stuff of science fiction. Instead, they are becoming indispensable partners for forensic professionals. This infusion of technology empowers experts to uncover truth with greater speed, precision, and confidence than ever before.

For example, fingerprints have been a gold standard in forensic identification for over a century. The traditional process involved a fingerprint examiner manually comparing a latent print from a crime scene to a suspect's known prints. This was a time-consuming and painstaking task. While databases like AFIS (Automated Fingerprint Identification System) helped, the final call always rested on human expertise. Modern forensic technology has supercharged this process. Artificial intelligence (AI) and machine learning algorithms can now analyze and compare millions of prints in seconds. These systems identify subtle patterns and minutiae that might be missed by the human eye. This does not replace the expert. Instead, it acts as a powerful assistant for the fingerprint analyst. The AI can quickly generate

a list of the most likely candidates from a massive database. The examiner then applies their expert judgment to confirm a match. Consequently, investigations move faster and the potential for error is significantly reduced.

Impression evidence, like shoe prints or tire tracks left in soil, provides valuable clues about events at a crime scene. For years, the standard collection method involved photography and making plaster casts. While effective, casts can be fragile and may miss microscopic details. Today, a forensic investigator can use portable 3D scanners to capture a perfect digital replica of the impression.

This high-resolution 3D model offers numerous advantages:

Permanent Preservation: The digital model is a permanent, non-degradable record of the evidence exactly as it was found.

Enhanced Analysis: A forensic analyst can manipulate the 3D model on a computer. They can change lighting angles to reveal tiny scratches, view the impression from any perspective, and take precise measurements of depth and wear patterns.

Clearer Courtroom Presentation: Presenting a dynamic 3D model to a jury is far more impactful than showing a static photograph or a physical cast.

This application of forensic technology allows for a far more detailed and objective analysis, strengthening the conclusions drawn from impression evidence.

The impact of forensic technology extends beyond individual disciplines. It is creating a more integrated and efficient workflow for all forensic scientists. A forensic investigator at a crime scene can use a mobile device to instantly upload digital images and 3D scans to a secure cloud server. Back in the lab, a team of analysts can access this data immediately and begin their work. This connectivity speeds up the entire investigative process. Furthermore, it ensures a clear and unbroken chain of custody for digital evidence. Every access and analysis step is logged automatically. This digital trail enhances the integrity of the evidence,

making it easier to defend in court. The seamless flow of information allows different experts to collaborate more effectively, piecing together a complete picture of the crime.

Despite its incredible benefits, the adoption of new forensic technology comes with challenges. Courts must be convinced of the scientific validity of these new methods. This requires extensive validation studies and clear explanations from forensic experts on the witness stand. There is also the risk of a new “CSI effect,” where juries expect flawless, instantaneous results from technology without understanding its limitations. The role of the human expert is not shrinking; it is evolving. Forensic scientists must now be proficient in both their core scientific discipline and the technology they use. They must be able to interpret the data generated by machines and explain its significance to a lay audience. The future will see an even deeper integration of AI and automation, but the critical thinking and ethical judgment of the human forensic specialist will remain essential.

8. Well Known Digital Evidence Tracking Systems

JusticeTrax

Based in Mesa, Arizona, JusticeTrax, Inc. a Versaterm Public Safety company, has been providing software to forensic science and law enforcement professionals since 1995.

It is used by many forensic labs to track evidence from intake to analysis. You can see every action, who handled the evidence, when and where.

It holds a strong chain of custody records; its barcode tracking reduces human error, and it helps labs pass audits and court scrutiny.

LabLynx

LabLynx, Inc. is a privately held American laboratory informatics company that develops, supports, and markets laboratory information management system (LIMS) solutions. It tracks samples and evidence in real time, stores full case data reports, and supports secure data sharing between agencies.

Its primary offerings over the years have included webLIMS and ELab. The company's primary clients include laboratories in the agriculture, clinical, environmental, forensics, health care, and manufacturing industries, including government agencies.

The company is known for introducing one of the first browser-based LIMS products in 1997 and being in the laboratory informatics industry for decades.

StarLIMS

StarLIMS is a Laboratory Information Management System (LIMS). It is used to manage and track laboratory data, samples, instruments, and workflows. It helps labs automate processes, manage resources, and ensure regulatory compliance. It is commonly used in industries such as pharmaceuticals, healthcare, food and beverage, and environmental testing.

Found in over 2,000 laboratories, STARLIMS serves Life Sciences, Contract Development Manufacturing Organizations (CDMOs), Food & Beverage, Chemical, Agrochemical, Oil & Gas, Consumer Goods, Contract Testing, R&D, Public Health, and Clinical Diagnostics organizations.

SAFE

SAFE, by Tracker Products, is a cloud-based evidence management solution designed to help police departments, state highway patrols, law firms, forensic labs, and other organizations manage the collection, processing, and storage of information. The platform enables users to track physical as well as digital assets via barcode scanning, e-signatures, and customizable dashboards.

SAFE's unbreakable chain of custody functionality allows institutions to track the movement, transfer, and storage of all items of evidence in their inventory. SAFE's key features include rolebased permissions, concurrent licensing, data import, barcode labeling, and SAML-based authentication.

SAFE comes with an auto-disposition module, which allows users to define follow-up dates for cases. Users can also automate notifications for upcoming review dates, information modifications, officers' responses, and outcomes.

FileOnQ

FileOnQ is a specialized, configurable software platform for public safety and law enforcement agencies designed to manage physical and digital evidence, assets, property, and crime scenes. It acts as a secure "evidence control center" to ensure chain of custody, streamline operations from intake to disposition, and reduce liability.

The system combines case management with evidence tracking, it has an easy access for officers and investigators and keeps full custody logs for court use.

9. The Use of Artificial Intelligence in Forensics

Digital forensics is defined by Interpol as a specialized branch of forensic science that focuses on electronic data. Digital forensic scientists are charged with extracting, showing, storing, analyzing and reporting digital data that may be relevant to an investigation, and the explosion of digital data in recent years has made it more challenging to complete these investigations. Many forensic experts believe that AI in digital forensics could redefine the industry, enhancing the efficiency and effectiveness of digital forensic investigations.

When completed manually by humans, evidence analysis can be difficult and time-consuming, and it also can be prone to human error. AI technology has the capability of automating analysis by gathering and interpreting large volumes of digital data, which can include emails, images, videos and more, giving investigators the opportunity to focus their efforts on the critical and creative thinking required during the analysis phase.

The fields of forensic and legal psychology are often used to better understand criminal behavior. Criminal profiling requires investigators to be experts in forensic and legal psychology, so that they can understand the nuanced behavior of criminals and the causes that may have triggered that criminal behavior.

Similar to digital forensics, AI technology is becoming a valuable partner in both of these fields, which rely heavily on data collection, interpretation and analysis to come to conclusions about a criminal's profile. Within these fields, investigators can use machine learning algorithms to analyze vast amounts of data. This ultimately uncovers previously unseen patterns that may offer additional insight into the behavior or actions of the individual.

Ethical and legal considerations of AI in forensics:

Artificial intelligence is the updated technology that everyone is talking about. It is redefining every industry around the world, and it can be exciting for those who see its potential. However, there are ethical and legal considerations to consider, particularly when it comes to AI and digital forensics. Currently, most forensics experts are only just beginning to understand the full power of AI technology and machine learning, and it's essential that all forensic scientists using this technology are transparent about it. Maintaining accountability and fairness is necessary, particularly because there is the risk of bias within the algorithms.

AI algorithms are incredibly complex, and most people who utilize everyday technology do not have a comprehensive understanding of the implications of machine learning. Digital forensic scientists relying on AI technology must be able to explain how the algorithms they are using have been developed, and subsequently, how they are being utilized within the context of their forensic investigations. Given the fact that many of these results are going to be shared as evidence within a courtroom, it's essential that forensic experts can explain the results of AI analysis in a way that can be understood both by lawyers and members of the public who are serving on a jury.

AI algorithms are based on human-created patterns, which means that bias can exist within these algorithms, just as they do within human nature. Acknowledging

the fact that AI bias can exist, particularly in areas such as facial recognition, risk assessment and criminal profiling, is essential. AI bias can be found within the training data that the algorithm uses, the algorithm itself or within the predictive analysis produced by the algorithm. Digital forensic scientists must work to develop and implement AI systems that are fair and unbiased to protect the integrity of their investigations and ensure that results are accurate.

Within a digital forensic investigation, protecting data privacy and enhancing security measures is critical. Given the fact that mass amounts of digital data are extracted and analyzed within a digital investigation, it's important to have strong data protection measures and ethical guidelines in place.

10. Ethical and Privacy Concerns in Forensics

Ethical dilemmas in forensic science

Professional credentials

These include misrepresentation of the credentials before the court of law. Misrepresentations include educational degree attainment (e.g. claiming an unearned Ph.D. or a degree was earned from a particular institution when, in fact, it was not), professional licensures or certifications (e.g. falsely claiming certification as a forensic Pathologist from the American Board of Pathology or a common tactic of equating having attained actual certification with being board "eligible"), employment history and data about previous testimonies such as number of times, locations, etc. Most often this is done to impress the client, the judge or jury to ward off the challenges such as cross-examination by exaggerating the qualifications. Due to lack of resources and fact-checking methods, such exaggerations are seldom caught, and such acts are unethical and must be dissuaded.

Laboratory analytical procedures

Most laboratories have validated and well-established protocols to be followed during tests of analysis as well as recording these tests and their results.

Laboratories place priority in the implementation of such protocols but, less than often, such protocols are not followed by the forensic scientist which is unethical. Unethical issues include making insufficient or indiscriminate analysis, and analysis to fit the written law. Sometimes forensic scientists report the results without even opening the containers; a practice known as 'dry-labbing'. Results and conclusions offered by forensic scientists must be explicit and clear.

Interpretation of analytical data and presentation of testimony in a court of law

In the courtroom, forensic scientists face many ethical dilemmas while providing their testimonies. Ethical dilemmas associated with the interpretation of analytical data and presentation of testimony in a court of law may include bias on the part of forensic scientists, use of scientific jargons, use of confusing or deceptive testimonies, excessive equivocacy, and advocacy. Another issue with forensic laboratories is the way results and conclusions are reported. Some laboratories report minimal results without any useful or appropriate explanations. Also, many a time, the forensic scientist who performed the analysis is not even required to be present in the court for the testimony.

Privately employed forensic scientists

An increasing number of private consultant practices posed a serious likelihood to the ethics in the field of forensic science. No disciplinary code can be applied to such private consultants. The professional integrity of a member is perhaps less of a problem in forensic science than in other professions, as we possess or perhaps suffer the most stringent form of quality control in the form of cross-examination in courts, where any malpractice, omissions or fudging is very likely to be revealed. The private consultancy offers the greatest risk of malpractice as there are less supervision, less peer review, and more financial incentive.

Publicly employed forensic scientists

In the case of public forensic laboratories, it must be noted that they are neither a part of nor administered by the local governments or local law enforcement agencies. This leads to the development of a usual perception that these laboratories are part of a local law enforcement agency. In such case, it is very necessary to maintain the autonomy of these laboratories to maintain a high ethical ground.

Obligations to the forensic science profession and professional skill maintenance

As a scientist, researcher and practitioner, all forensic scientists have the innate responsibility and obligation towards the forensic science profession to maintain the higher ethical values and standards. Ethical dilemmas include three categories that are a failure to keep up to date with recent advancements and updated knowledge, improper use of proficiency tests, continuing the improper educational practice.

Objectivity in forensic science

Applebaum (1997) pointed in his paper on ethics in forensic psychiatry that the basic principle of ethics is telling the truth and distinguishes between subjective and objective truth telling. Subjective truth telling is to state what we believe is true whereas the objective is to recognize the limitations of methods used to reach conclusions. It includes recognizing the limitations of our scientific and professional knowledge which has led to deduction of conclusions. It is objective to include literature in reports which support, as well as contradicts our conclusions in order to use the explanatory framework that is widely accepted by the scientific community. In his treatise on general psychotherapy, Jasper (1997) differentiated between the objective and subjective phenomena; he identified the objective as any trained observer where they are perceived by our senses whereas the subjective cannot be perceived by sense organs. Norko (2005) stated that the subjective element of truth telling involves the expert's honesty whereas the objective element relates to his knowledge and testimony.

Similarly, forensic scientists must remain objective while reaching conclusions that can be attained through training and following a standard ethical code. The ethical forensic scientist is a scientist who strives to reach conclusions based on examinations performed without any bias or extending themselves beyond their capabilities or talents (Murdock and Holmes 1991). They must not forget that objectivity is their main attribute and that they must examine all the angles before reaching conclusion. They have responsibilities towards the public, therefore, their examinations and analyses must be accountable and objective (Murdock and Holmes 1991).

Ethical dilemmas in forensic genetics

Collection and storage of DNA samples

Obtaining samples for DNA database is one of the biggest and most debated upon issues. The first issue is who will be providing samples in respect to the criminal investigation? The second issue is who will be required to provide a sample, the profile from which will be stored in DNA database?

The UK National DNA database was first forensic DNA database established in 1995. Its expansion to include DNA profiles of millions of innocent citizens into the database was widely criticized for it was considered as a breach of personal space. In 2008, the Grand Chamber of the European Court of Human Rights in the case of *Marper vs. the UK* reached a unanimous judgment that the indefinite retention of innocent people's DNA profiles, fingerprints and samples breached privacy (Wallace et al. 2014). Different countries have different criteria to whose DNA sample must be obtained. For instance, in Australia, samples of those who are suspected of having committed crimes for which the penalty is more than 5 years are taken. Samples and profile must be destroyed within 12 months if the charge sheets are not filed, the prosecution is abandoned, or the accused is acquitted. On the other hand, if the accused has been convicted, his profile is retained in the database. In the UK, the profiles of all the accused of recordable crimes that include offenses for which there is a jail term if convicted, are recorded in the database. In France, the database is reserved only for those accused of sexual

crimes. An audit system to monitor and prevent the unethical use of such databases is also vital.

Testing the samples/ using the results

Correct sample collection, security, transport, storage along with processing and analysis are important conditions required for the high-quality database management. Meeting these conditions results in high level of confidence which in turn results in high reliability and high credibility. Contamination problems are a severe predicament and strong reason for doubtful credibility. Contamination can occur at various steps of analytical procedures such as sample collection, securing, transport, storage, and analysis. Although contaminations can never be prevented, it is imperative that they must be minimized and estimation errors are calculated.

Access and retention of DNA samples

DNA profiles are stored in databases. Many authors claim the correlation of DNA markers with the physical characteristics. Moreover, DNA might reveal illness history. In pursuit of establishing such correlations and investigating them, researcher tends to overuse these databases. This will further put pressure on other databases to provide information for researcher's assistance.

Misuse of genetic research in application of genetics to forensic sciences

One of the discussed topics relate to the ethnic and racial labels to the genetic samples. Often forensic scientists try to use results from genetic research to put ethnic and racial labels on the samples encountered on the crime scene. However, many authors believe that the ethnic and racial differences are cultural in nature

rather than biological or genetic. Many have raised questions on the scientific utility of racial classifications.

As a conclusion, scientists and researchers are always baffled by the role of ethics. In multiple situations, ethics and research cross the path. Ethics are the soul of any profession and without it, the meaning of profession is vague and ambiguous. Ethics also help in establishing quality, validity, and authenticity of the profession. Although what is ethical to one person may be unethical to another, principles of ethics must be followed. Forensic science deals with the legal aspects and may help in establishing the guilt or exonerating the accused. Therefore, it is mandatory for every forensic organization to have an ethical code which guides forensic scientists to perform their duty with honesty and passion. Definition and limits of following ethical guidelines may vary from person to person but the minimum set of ethics must be made mandatory to be followed especially in the field of forensic science.

11.Past Actions

a. Major Key Moments in the Forensic Science History

1. 1814 – toxicology

Spanish-born French toxicologist and chemist Mathieu Orfila published the first comprehensive textbook on poisons, *Traité des Poisons*, in two volumes, from 1814, and which was translated in English three years later. In a later edition, he categorised toxins into four groups for the first time: irritants, narcotics, acrid narcotics, which was a combination of the two previous categories, and septic poisons. Orfila's work came to the fore in 1840, when he was called on to apply his chemical knowledge to the investigation into Marie Lafarge who was accused of having killed her husband with arsenic.

2. 1835 – ballistics

The first documented use of ballistics in crime investigation was by Bow Street Runner Henry Goddard in a case of murder in Southampton in 1835. However, this does not mean it was the first case ever, as there may well have been other, earlier investigations in which bullets from a murder victim were compared with the mould in which they were cast, especially as these were individual and unique.

3. 1836 – chemical analysis

In one of the earliest major advances in modern toxicology, British chemist James Marsh developed his famous test for arsenic, which was first used in the investigation into the murder of George Bodle in 1833. There had been rudimentary tests – based on reduction and precipitates – before this time, but Marsh's experiment became the standard process. After the trial, Marsh refined the test and the results were published for the first time in the *Edinburgh New Philosophical Journal*, in 1836. Marsh's test, which could detect the presence of arsenic as well as the quantity, was modified over the decades until the 1970s.

4. 1861 – hair analysis

Polish pathologist Rudolf Virchow was the first person to use hair analysis from samples collected from a victim in a crime case in 1861. He discovered that the characteristics of hair in individuals varies, but also recognised its limitations – he concluded that the hair of individuals can be similar, thus undermining the relevance of hair analysis in legal trials. Known as the 'father of pathology,' Virchow made other significant contributions to medical science – he was the first person to develop a standardised procedure for autopsies and to correctly link cancer to changes in normal cells. In 1847, he identified leukaemia.

5. 1877 – fingerprinting

Sir William Herschel, an English magistrate in India formally proposed using fingerprints for identifying criminals in 1877 – he had used a handprint for the first time in 1858, to prevent fraud in a building supplies contract. Herschel, along with Dr Henry Faulds, a physician working in Tokyo, set in motion a quest to develop this scientific technique which, towards the end of the nineteenth century, would see its inception as a forensic tool in crime investigations. From the 1870s, the history of fingerprinting becomes complex as individual rivals sought to beat the competition to categorise and standardise fingerprint analysis. In 1901, the scientific race culminated in the introduction of the Fingerprint Bureau in the Metropolitan Police by Sir Edward Henry.

6. 1892 – fingerprint analysis

The first ever homicide solved using fingerprint analysis took place in Argentina in 1892, when two children were found murdered at their home in the province of Buenos Aires – their mother had been injured in the attack but survived. The previous year, police criminologist Juan Vucetich had established a fingerprint database in the police department in La Plata. He was called to investigate this complex case, in which a neighbour was the prime suspect.

Vucetich analysed bloody fingerprints found at the crime scene against the children's mother's and discovered that they were a match. Francisca Rojas then confessed to their murder. In Britain, the first murder solved by fingerprint analysis was seven years later, in 1905.

7. 1893 – CSI

Austrian jurist and criminologist Hans Gross published the first significant textbook on applying science to criminal investigations in 1893. It was translated into English in 1906 and republished as *Criminal Investigation: A Practical Handbook*. This pioneering manual presented new tools, processes and behaviours for investigators, and created the 'crime scene' with the focus on the location of the victim, rather than on the body itself. This was the first time that forensic science had been formally applied as a tool in crime investigation, providing a clear link

between scientific criminology and forensic science. Gross also developed a system of management and preservation of crime scene evidence, which helped police detectives to assess a murder more efficiently and effectively.

8. 1901 – blood analysis

One of the most significant advances in forensic science was when Paul Uhlenhuth, a German immunologist successfully developed a chemical test to distinguish human from animal blood in 1901. This was a precipitin test which used specific antibodies to bind with soluble antigens, which can be visualised under a microscope. From then on, the test was increasingly used in homicide investigations. In the same year, Austrian American biologist and Nobel Prize winner, Karl Landsteiner identified the three blood groups, A, B and O – ‘O’ was originally referred only to as ‘C’ – by mixing red blood cells and serum. He discovered that the serum from some individuals led to the red blood cells clumping together. This led to the first successful blood transfusion six years later. In relation to crime investigation, this forensic technique enabled investigators to identify the type of blood found at crime scenes for the first time.

9. 1910 – cross contamination

The concept of cross-contamination at crime scenes was developed by French scientist Edmond

Locard, who instigated the first ever forensic police laboratory in Lyon in 1910. Locard’s Exchange Principle established the premise that ‘every contact leaves a trace,’ in that a perpetrator leaves organic traces at a crime scene and takes others away with them. It is not known exactly when Locard first used the phrase, and it is more likely that it evolved over time through his published work. However, it remains the basis of crime scene investigation and the collection and preservation of evidence today. Also, Locard’s Exchange Principle is often referred to in crime fiction books and TV shows.

10. 1984 – DNA profiling

This groundbreaking forensic science technique was discovered by British geneticist Sir Alec Jeffreys at the University of Leicester. It was first used, two years later, during the investigation into the murders of Lynda Mann and Dawn Ashworth, and it led to a successful conviction after genetic profiles were created of men in the area where the victims had lived.

12. What a Resolution Should Address

1. How can international cooperation ensure that forensic data is shared effectively while still protecting individual privacy and maintaining high ethical standards?
2. How can the integration of Artificial Intelligence improve the speed and accuracy of evidence collection while reducing the risk of human error at crime scenes?
3. How can member states harmonize their legal frameworks to ensure that evidence tampering is met with consistent penalties and robust enforcement across international borders?
4. How can member states institutionalize ethical standards in forensic science to ensure long term compliance and professional accountability within their national justice systems?
5. What safeguards must be in place to ensure that forensic evidence is not fabricated or 'planted' by law enforcement agencies to target political dissidents or marginalized groups?
6. What specific international penalties or reporting mechanisms should be set up to address forensic spoliation by state or non-state actors?

7. What mechanisms can be implemented to ensure that developing nations have access to the technological infrastructure required to meet international standards?
8. What international rules should be set to protect the digital chain of custody and ensure that evidence from the cloud is not changed or tampered with?
9. How can member states and international agencies like INTERPOL improve their cooperation to ensure that forensic evidence is shared quickly and follows a secure chain of custody across borders?
10. How can we build a 'global circle of trust' so that forensic evidence collected by one agency is automatically accepted as authentic and untampered with by another country's legal system?

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